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Plan of Internship program

Research Rummer Internship IIT Palakkad - 45days [6 weeks 3/6/2024 - 16/7/2024]					
Topic	Power Electronic Circuits - Numerical Simulations			Dept	EE - Electrical Engineering
				Faculty	Dr. Naga Brahmdra - Assistant Professor
Weeks	Details			Weeks	Details
Week - 1	Introduction to power electronics. Software setup[LT spice, Altium Pcb design]. Voltage pulse generator and MOSFET charactries. Power electronics converters _ circuit practice[12 Examples] [Learnings - Converters and EDA Tools]			Week - 4	NCP 51820 Gate Driver _ ON semi GaN 3R2_100 CBE _ MOSFET _ Nexperia Data sheet, Symbol and Component creation in LT Spice and Altium PCB design [learnings - Creating of Libraries in EDA Tools and Gate driver and MOSFET]
Week - 2	Power Electronics Circuits design and Numarical calculations and creating a formula sheet and Simulating in LT spice tool. 1.Buck 2.Boost and 3.Buck Boost [Learnings - Calculation and circuit design and simulations in LTspice]			Week - 5	Design and Simulating of Synchronous Buck boost converter Schmetic design and PCB layout component assembly and routing [Learnings - Altium PCB desining]
Week - 3	Power Electronics Circuits design and Numarical calculations and creating a formula sheet and Simulating in LT spice tool. 1.Forward [CCM & DCM] 2. Flyback 3.Pushpull [Learnings - Calculation and circuit design and simulations in LTspice]			Week - 6	Evaluation board design refinement and design rule checks Finalize evaluation board design Prepare documentation for evaluation board Review and consolidate all internship learnings Final project presentation preparation Final project presentation and internship wrap-up

Out comes
* Power Electronics Converters * Calcuation and formula based sheets creation * Circuit Simulation in LT Spice Tool * Gate Drivers _ NCP51820 _ Onsemi * GaN MOSFET GAN3R2-100CBE _ Nexperia * EDA Tools - Altium PCB design * Skills - LT spice simulations, Numerical calculations and Altium Schematic and PCB designing.